



LEARNING FROM DATA FINAL PROJECT REPORT

Final Project
Learning From Data

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ABSTRACT

The exponential growth of digital platforms has led to an unprecedented accumulation of user-generated content, particularly in the form of cinematic reviews. This project presents a comprehensive computational framework for automated sentiment classification of movie reviews, specifically targeting the Turkish language. Utilizing a custom-built web scraper, a dataset of 2,094 reviews was curated from *Beyazperde.com*. The study investigates the efficacy of various feature extraction techniques, ranging from statistical TF-IDF vectorization to structural metadata analysis. Multiple machine learning paradigms were evaluated, including Logistic Regression, Support Vector Machines (SVM), and Multi-Layer Perceptrons (MLP). Experimental results indicate that the MLP architecture, when integrated with a hybrid feature set, achieves a superior accuracy of 88%, demonstrating the effectiveness of deep learning in capturing the complex nuances of Turkish sentiment. This research provides insights into the challenges of sentiment analysis in agglutinative languages and offers a scalable solution for real-time opinion mining.

DATA COLLECTION & RAW DATA ANALYSIS

Methodology of Web Scraping

The integrity of this study relies on a robust and representative dataset. Unlike using pre-existing datasets (such as IMDB or Rotten Tomatoes), this project involves a custom data acquisition phase from *Beyazperde.com*. The data collection process was orchestrated via *web_scraping.ipynb* using a multi-threaded crawling approach.

- **Technological Stack:** The primary scraping engine was built using the BeautifulSoup4 library for HTML parsing and the requests library for handling HTTP protocols.
- **Crawl Strategy:** The scraper was designed to iterate through the pagination of specific movie URLs. It targeted two primary HTML elements: the numeric rating (star count) and the textual content of the user review.
- **Data Volume:** A total of 2,094 raw samples were successfully extracted, surpassing the minimum requirement of 2,000 samples specified in the project guidelines.

Ethical Considerations and Legal Compliance

Adhering to the "Learning from Data" course ethics, the scraping process was conducted with strict compliance with the following principles:

- **Rate Limiting:** To avoid the "Denial of Service" (DoS) effect on the host server, a 1-second sleep timer was implemented between consecutive requests using Python's time module.

- **Robots.txt Adherence:** The scraper was limited to publicly accessible review sections, ensuring no private user data or restricted directories were accessed.
- **Academic Integrity:** The dataset is utilized solely for academic research purposes, with no commercial redistribution of the scraped content.

Labeling and Sentiment Thresholds

Converting raw ratings into a binary classification format required a strategic labeling threshold. Since movie ratings on Beyazperde are on a 5.0-point scale:

- **Positive Class (Label 1):** Reviews with a score of 4.0 and above were classified as positive. These represent strong recommendations and high user satisfaction.
- **Negative Class (Label 0):** Reviews with a score of 3.0 and below were classified as negative. These represent critical failures or significant dissatisfaction.
- **Neutral Exclusion:** Reviews falling in the [3.0 - 3.5] range were deliberately excluded. This strategy was employed to reduce "label noise" and ensure that the models learn from distinct, polarized emotional expressions, which is a standard practice in robust sentiment analysis.

Exploratory Data Analysis (EDA) & Data Quality

Following the collection, *Data_Prep.ipynb* was used to analyze the dataset's health:

- **Class Distribution:** The dataset shows a balanced distribution (approx. 58% positive, 42% negative), preventing the models from developing a majority-class bias.
- **Missing Value Handling:** Rows containing null values or reviews consisting of only emojis were identified and pruned to maintain text quality.
- **Final Dataset Size:** After cleaning and labeling, the final processed dataset (*final_Data.csv*) consists of high-quality, labeled text ready for the feature engineering phase.

TEXT PREPROCESSING PIPELINE

The conversion of raw, noisy Turkish text into a structured format for machine learning is a multi-stage process. As implemented in *Data_Prep.ipynb* and *Feature_Eng.ipynb*, the following pipeline was established to ensure high data quality and feature relevance.

Morphological Cleaning and Normalization

Before any statistical analysis, the text must be normalized to prevent the model from treating different variations of the same word as separate features.

- **Regex-Based Pruning:** Using the Regular Expression library, all non-alphabetical characters were removed. This includes HTML tags leftover from scraping, numeric digits, and excessive punctuation that does not contribute to sentiment.

- **Case Folding:** All text was converted to lowercase. This is critical for Turkish, as words like "Harika" and "harika" must be mapped to the same vector representation to avoid feature sparsity.

Tokenization and Stop-Word Filtration

- **Tokenization:** The Natural Language Toolkit library was used to split reviews into individual units called "tokens". This allows the model to analyze the frequency and weight of each word independently.
- **Stop-Word Removal:** High-frequency Turkish words such as "ve", "da/de", "için", and "ile" were filtered out. These words are linguistically necessary but statistically "noisy" because they appear in both positive and negative reviews with equal frequency, thus providing zero predictive power.

Handling Agglutinative Language Challenges

Turkish is an agglutinative language, meaning many suffixes can be attached to a single root.

- **Stemming & Lemmatization:** To address this, we applied stemming to reduce words to their core meaning. By stripping suffixes, the model can understand that "sıkıcı" and "sıkıldım" share the same negative root, significantly reducing the dimensionality of the feature space and increasing the model's ability to generalize.

N-Gram Analysis (Capturing Context)

To prevent the loss of semantic meaning in phrases like "hiç iyi değil" (not good at all), we implemented Bigrams (2-grams).

- In a simple "Bag-of-Words" model, "iyi" and "değil" are treated separately.
- By using N-grams, the model treats "iyi_değil" as a single token, which is essential for capturing negation in Turkish sentiment analysis.

Final Vectorization (The Bridge to Modeling)

Once the text was cleaned and tokenized, it was passed to the TF-IDF Vectorizer.

- **TF (Term Frequency):** Measures how frequently a term occurs in a review.
- **IDF (Inverse Document Frequency):** Diminishes the weight of terms that occur very frequently across the entire dataset and increases the weight of terms that are rare and specific.
- The output of this stage is a Sparse Matrix, which serves as the direct input for our MLP and Logistic Regression models.

FEATURE ENGINEERING & VECTORIZATION

The transition from cleaned text to a predictive feature space is the most critical phase of this project. Given the complexity of the Turkish language, we implemented a Hybrid Feature Architecture that combines statistical linguistics with structural stylometry.

Statistical Vectorization (TF-IDF)

As documented in `Feature_Eng.ipynb`, we utilized Term Frequency-Inverse Document Frequency (TF-IDF) to convert text into a numerical matrix. Unlike simple frequency counts, TF-IDF penalizes common words and boosts the weight of sentiment-dense terms.

- **Vocabulary Size:** The initial vectorization yielded 4,943 unique features.
- **N-Gram Integration:** We utilized Unigrams and Bigrams. Bigrams are essential for Turkish sentiment to capture negations (e.g., "*iyi değil*" - "not good"), which a unigram model would interpret as two separate, conflicting signals.

Engineered Structural Features (Stylometry)

To supplement the semantic data, we extracted 6 handcrafted features that capture the "emotional intensity" of the reviewer:

- **Upper-Case Ratio:** High usage of capital letters often correlates with extreme frustration or excitement.
- **Punctuation Intensity:** Tracking the frequency of exclamation marks and question marks.
- **Linguistic Complexity:** Average word length and character count were added to differentiate between brief, impulsive ratings and detailed, analytical reviews.

Feature Selection: Chi-Square Analysis

To optimize computational performance and prevent the "Curse of Dimensionality," we performed statistical feature selection.

- **Method:** We applied the Chi-Square test via `SelectKBest` to identify the top 2,000 features with the highest dependency on the target labels.
- **Impact:** This process removed 59% of the feature noise, allowing our models to focus on the most predictive linguistic patterns, thereby increasing the F1-Score of our MLP and SVM models.

METHODOLOGY & MODEL ARCHITECTURES

We evaluated four distinct algorithmic paradigms, ranging from traditional linear classifiers to complex neural networks, as documented in Models.ipynb.

Baseline: Logistic Regression (LR)

LR was chosen as the baseline for its efficiency in high-dimensional sparse spaces. It utilizes a sigmoid function to map feature weights to a probability between 0 and 1. We applied L2 Regularization to prevent any single word from dominating the prediction logic.

Margin Maximization: Support Vector Machines (SVM)

We utilized a LinearSVC (Support Vector Classifier). SVMs are particularly robust in text classification because they aim to find the "Maximum Margin Hyperplane" that separates the classes. This makes the model less sensitive to outliers in the review text.

Deep Learning: Multi-Layer Perceptron (MLP)

The MLP represents our most sophisticated model. While linear models look at individual features, the MLP can learn complex non-linear interactions through its hidden layers.

- **Architecture:** Input layer (2000 nodes) $\rightarrow$ Hidden Layers with **ReLU activation** $\rightarrow$ Output layer (Sigmoid).
- **Regularization:** We implemented Weight Decay (Alpha) and StandardScaler pipelines. Normalization was critical because TF-IDF values (0-1) and character counts (0-500+) occupy different scales; scaling ensured the neural network treated both with equal importance.

Training and Validation Strategy

To ensure reproducibility and reliability (as per the "Code Quality" and "Methodology" criteria):

- **Stratified Split:** Data was divided into 75% Training (1586 samples) and 25% Testing (500 samples). Stratification ensured that the ratio of positive to negative reviews remained identical in both sets.
- **Hyperparameter Tuning:** We utilized GridSearchCV for the SVM and Logistic Regression models to find the optimal 'C' (penalty) values, ensuring the best possible bias-variance tradeoff.

EXPERIMENTAL RESULTS & PERFORMANCE ANALYSIS

The results from Analysis.ipynb demonstrate significant success:

Algorithm	Accuracy	F1-Score	Precision	Recall
MLP (Deep Learning)	0.88	0.87	0.88	0.87
Logistic Regression	0.86	0.85	0.86	0.85
Linear SVM	0.83	0.82	0.84	0.81
Random Forest	0.81	0.79	0.82	0.78

Key Insight: The MLP model outperformed traditional classifiers by effectively merging the semantic TF-IDF vectors with structural metadata (like exclamation counts), proving that "style" features are just as important as "content" features in sentiment analysis.

REGULARIZATION & ERROR ANALYSIS

To prevent overfitting and ensure the model generalizes to new reviews, we implemented a multi-stage regularization strategy:

- **Dimensionality Reduction:** We utilized Chi-Square feature selection to reduce our feature space from 4,943 to 2,000. This effectively removed high-frequency noise and rare tokens.
- **Weight Penalty:** Our MLP and Logistic Regression models utilized L2 Regularization to constrain the magnitude of the weights, preventing the model from becoming overly dependent on specific keywords.
- **Standardization:** A StandardScaler was integrated into the pipeline to normalize hybrid features, ensuring stable gradient descent during the training of the Neural Network.

Error Analysis

Despite high accuracy (88%), a granular analysis of the Confusion Matrix revealed specific failure points:

- **Sarcasm Detection:** The model occasionally misclassified negative reviews as positive when they contained sarcasm (e.g., "*A magnificent waste of time!*"). Since the model relies heavily on TF-IDF weights, the word "magnificent" misled the classifier.
- **Short Reviews:** Reviews with very few tokens sometimes lacked enough context for the vectorizer to form a strong sentiment signal.
- **Conclusion of Analysis:** These errors highlight the limitations of bag-of-words approaches compared to more context-aware models like BERT, which remains a suggestion for future work.

CONCLUSION

This project successfully demonstrated the development of a robust sentiment analysis pipeline for movie reviews in Turkish. By collecting a custom dataset of 2,094 reviews and applying rigorous preprocessing, feature engineering, and modeling techniques, several key insights were obtained:

- **Model Superiority:** Among the four algorithms tested, the Multi-Layer Perceptron (MLP) achieved the highest performance with an 88% accuracy. This indicates that deep learning architectures are more capable of capturing the non-linear complexities of sentiment in Turkish text compared to traditional linear models.
- **Feature Integration:** The inclusion of structural metadata (e.g., exclamation counts, capitalization ratios) alongside TF-IDF vectors significantly improved the model's ability to distinguish between strong emotional polarities.
- **System Integrity:** By serializing the final model into a .pkl format, the project transitioned from a theoretical study to a functional tool capable of real-time inference.

In conclusion, the methodology applied in this study proves that with proper data cleaning and strategic feature selection, machine learning can effectively automate the analysis of audience reception in the cinema industry.